Arista Access Node KR-ACC-24 Routing Protocol Flap Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_2523

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Node ID KR-ACC-24, a member of the KR Puram Access site's access network, which is comprised of Arista networking equipment operating within the context of an OSPF-based IGP. The root cause of this failure domain is attributed to an unstable BGP routing policy, OSPF adjacency instability, or ISIS LSP flooding, which can manifest as a result of misconfigured routing policies, inadequate ACLs, or hardware issues. This SOP is designed to guide the troubleshooting and resolution process, ensuring timely and effective restoration of the affected network.

Pre-requisite Checks

- Check: Verify the site's environmental conditions, including temperature and humidity levels, to ensure they are within manufacturer's specified ranges., Task: Dispatch remote hands to KR Puram Access to perform site checks and verify redundant link status.

- Check: Verify the node's power supply and ensure it is properly connected to a reliable power source., Task: Check the node's console and out-of-band management interfaces for any errors or issues.

Potential Root Causes

Cause	Probability
Unstable BGP routing policy	High
OSPf adjacency instability	Medium
ISIS LSP flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command provides an overview of the BGP session's state, including the number of established sessions, total prefixes advertised, and the number of routes received.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about the OSPF neighbors, including their state, IP address, and interface.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays the ISIS database, including the level of the LSPs, sequence numbers, and the sequence number of the last LSP received.
```

Resolution Steps

Step 1: Step: Verify and adjust BGP routing policy, Command: show bgp summary | i established, Description: Verify the BGP session's state and adjust the routing policy as necessary to stabilize the BGP session.

Step 2: Step: Verify OSPF adjacency, Command: show ospf neighbor | ex up, Description: Verify the OSPF adjacency state and re-establish the adjacency as necessary.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis database, Description: Clear the ISIS database to resolve LSP flooding issues.

Post-Check Verification

```
Command: show bgp summary | i established, Expected Output: BGP session established, Description: Verify the BGP session's state to ensure it is established.

Command: show ospf neighbor | ex up, Expected Output: OSPF adjacency established, Description: Verify the OSPF adjacency state to ensure it is established.

Command: show isis database, Expected Output: ISIS database stable, Description: Verify the ISIS database to ensure it is stable and free of LSP flooding.
```

Roll-back Procedures

- Step: Restore previous BGP routing policy, Command: show bgp summary | i established, Description: Restore the previous BGP routing policy to its original configuration if the resolution fails.

- Step: Restore previous OSPF adjacency, Command: show ospf neighbor | ex up, Description: Restore the previous OSPF adjacency configuration if the resolution fails.

- Step: Restore previous ISIS database, Command: show isis database, Description: Restore the previous ISIS database configuration if the resolution fails.